



Pimpri Chinchwad Education Trust's
Pimpri Chinchwad College of Engineering (PCCoE)

(An Autonomous Institute)

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TY B.Tech

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Assignment 4

Title: Create an Interactive Data Visualization Dashboard for a Real-Life Application (e.g., Weather Dashboard) using Charts and Graphs (Chart.js / D3.js)

Objectives:

- To understand the concept of data visualization in web applications.
- To design an interactive dashboard for real-life data representation.
- To visualize data using charts and graphs with libraries such as Chart.js or D3.js.
- To learn how JavaScript integrates with visualization libraries.
- To present complex data in an easy-to-understand graphical format.

Outcomes:

- Students will be able to create interactive dashboards using visualization libraries.
- Students will understand how graphical representation improves data analysis.
- Students will gain experience integrating external JavaScript libraries into web applications.
- Students will learn how to display real-world datasets using charts such as bar, line, and pie charts.

Theory Concept:

A **Data Visualization Dashboard** is a web-based interface that displays information visually using charts, graphs, and indicators. Instead of viewing raw numerical data,

users can easily analyze trends, comparisons, and patterns through graphical representation.

1. Data Visualization

Data visualization is the process of representing data using visual elements such as charts, graphs, and maps. It helps users quickly understand large datasets and make informed decisions.

2. HTML (Structure)

HTML is used to create the layout of the dashboard including containers, chart sections, headings, and data panels.

3. CSS & Bootstrap (Styling and Responsiveness)

CSS and Bootstrap are used to design visually appealing dashboards. Bootstrap provides responsive grids, cards, and layout utilities that ensure compatibility across devices.

4. JavaScript (Interactivity)

JavaScript handles dynamic behavior such as:

- Loading data
- Updating charts
- Handling user interaction
- Changing datasets dynamically
-

5. Chart.js / D3.js Libraries

Visualization libraries simplify chart creation:

- **Chart.js:** Easy-to-use library for creating responsive charts like bar, line, pie, and doughnut charts.
- **D3.js:** Advanced visualization library used for custom and highly interactive visualizations.

These libraries convert numerical data into graphical formats using JavaScript.

6. Real-Life Application – Weather Dashboard

A weather dashboard visually represents weather-related data such as:

- Temperature trends
- Humidity levels
- Rainfall statistics
- Weekly forecasts

Charts help users easily analyze weather patterns instead of reading raw data values.

Dataset Used:

The dashboard uses **sample or real-world weather data**, which may include:

- Temperature readings (daily/weekly)
- Humidity percentage
- Wind speed
- Rainfall data
- Weather conditions

Code:

Index.html

```
<!DOCTYPE html>
<html lang="en">

<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Weather Analytics Dashboard</title>
  <!-- Fonts -->
  <link rel="preconnect" href="https://fonts.googleapis.com">
  <link rel="preconnect" href="https://fonts.gstatic.com" crossorigin>
  <link
href="https://fonts.googleapis.com/css2?family=Outfit:wght@300;400;500;600;700&disp
lay=swap" rel="stylesheet">
  <!-- Icons -->
  <link rel="stylesheet"
href="https://cdnjs.cloudflare.com/ajax/libs/font-awesome/6.4.0/css/all.min.css">
  <!-- Styles -->
  <link rel="stylesheet" href="style.css">
  <!-- Chart.js -->
  <script src="https://cdn.jsdelivr.net/npm/chart.js"></script>
</head>

<body>
  <div class="app-container">
    <!-- Sidebar -->
    <aside class="sidebar">
      <div class="logo">
        <i class="fa-solid fa-cloud-bolt"></i>
        <span>Aether</span>
      </div>
      <nav>
        <a href="#" class="active" data-target="all"><i class="fa-solid
fa-house"></i> Dashboard</a>
        <a href="#" data-target="temp"><i class="fa-solid
fa-temperature-half"></i> Temperature</a>
        <a href="#" data-target="uv"><i class="fa-solid fa-sun"></i> UV
Index</a>
```

```

        <a href="#" data-target="wind"><i class="fa-solid fa-wind"></i>
Wind</a>

    </nav>

    <div class="user-profile">
        

        <div class="user-info">
            <h4>Paritosh</h4>
            <p>Free Plan</p>
        </div>
    </div>
</aside>

<!-- Main Content -->
<main class="main-content">
    <header>
        <div class="search-bar">
            <i class="fa-solid fa-location-dot"></i>
            <select id="citySelect">
                <option value="Pune">Pune, Maharashtra</option>
                <option value="Mumbai">Mumbai, Maharashtra</option>
                <option value="Delhi">New Delhi, Delhi</option>
                <option value="Bangalore">Bangalore, Karnataka</option>
            </select>
        </div>
        <div class="header-actions">
            <button class="icon-btn"><i class="fa-regular
fa-bell"></i></button>
            <button class="theme-toggle"><i class="fa-solid
fa-moon"></i></button>
        </div>
    </header>

    <div class="dashboard-grid">
        <!-- Current Weather Card -->
        <div class="card weather-card big-card" data-tab="all">
            <div class="weather-info">
                <div class="location">
                    <h2>San Francisco</h2>
                    <p>California, PN</p>
                </div>
                <div class="temp-display">
                    <span class="temp">22°</span>

```



```

        </div>
        <p class="text-secondary">Direction: NE</p>
    </div>
</div>

<!-- Temperature Trend Chart -->
<div class="card chart-card wide-card" data-tab="temp">
    <div class="card-header">
        <h3>Temperature Forecast</h3>
        <select class="time-range">
            <option>Weekly</option>
            <option>Monthly</option>
        </select>
    </div>
    <div class="chart-container">
        <canvas id="tempChart"></canvas>
    </div>
</div>

<!-- Chance of Rain -->
<div class="card rainfall-card" data-tab="rain">
    <h3>Chance of Rain</h3>
    <div class="chart-container">
        <canvas id="rainChart"></canvas>
    </div>
</div>

<!-- 7 Day Forecast List -->
<div class="card forecast-card tall-card" data-tab="forecast">
    <h3>7-Day Forecast</h3>
    <ul class="forecast-list">
        <!-- Populated by JS -->
    </ul>
</div>
</div>
</main>
</div>
<script src="script.js"></script>
</body>

</html>

```

Style.css

```
:root {
  --bg: #0f172a;
  --panel: #1e293b;
  --text: #f8fafc;
  --muted: #94a3b8;
  --accent: #38bdf8;
  --radius: 12px;
  --border: 1px solid rgba(255, 255, 255, 0.07);
}

* {
  margin: 0;
  padding: 0;
  box-sizing: border-box;
}

body {
  display: flex;
  min-height: 100vh;
  font-family: 'Outfit', sans-serif;
  background: var(--bg);
  color: var(--text);
}

/* Sidebar */
.sidebar {
  width: 220px;
  background: var(--panel);
  padding: 2rem 1rem;
  position: fixed;
  height: 100%;
  display: flex;
  flex-direction: column;
  border-right: var(--border);
}

.logo {
  display: flex;
  align-items: center;
  gap: 0.75rem;
  font-size: 1.3rem;
  font-weight: 700;
  color: var(--accent);
}
```

```
    margin-bottom: 2rem;
}

.sidebar nav {
  display: flex;
  flex-direction: column;
  gap: 0.5rem;
  flex: 1;
}

.sidebar nav a {
  text-decoration: none;
  color: var(--muted);
  padding: 0.75rem 1rem;
  border-radius: var(--radius);
  display: flex;
  align-items: center;
  gap: 0.75rem;
  transition: background 0.2s, color 0.2s;
}

.sidebar nav a:hover,
.sidebar nav a.active {
  background: rgba(56, 189, 248, 0.1);
  color: var(--accent);
}

.user-profile {
  display: flex;
  align-items: center;
  gap: 0.75rem;
  padding-top: 1.5rem;
  border-top: var(--border);
}

.user-profile img {
  width: 36px;
  height: 36px;
  border-radius: 50%;
}

.user-info h4 {
  font-size: 0.85rem;
}
```

```
.user-info p {
  font-size: 0.75rem;
  color: var(--muted);
}

/* Main */
.main-content {
  margin-left: 220px;
  padding: 2rem;
  width: 100%;
}

header {
  display: flex;
  justify-content: space-between;
  align-items: center;
  margin-bottom: 1.5rem;
}

.search-bar {
  display: flex;
  align-items: center;
  gap: 0.75rem;
  background: var(--panel);
  padding: 0.6rem 1.2rem;
  border-radius: 30px;
  border: var(--border);
  width: 360px;
}

.search-bar select {
  background: transparent;
  border: none;
  color: var(--text);
  outline: none;
  font-family: inherit;
  font-size: 0.95rem;
  width: 100%;
  cursor: pointer;
}

.search-bar select option {
  background: var(--panel);
}
```

```
.header-actions {
  display: flex;
  gap: 0.75rem;
}

.icon-btn,
.theme-toggle {
  background: var(--panel);
  border: var(--border);
  color: var(--text);
  width: 38px;
  height: 38px;
  border-radius: 50%;
  cursor: pointer;
  display: flex;
  align-items: center;
  justify-content: center;
  transition: color 0.2s;
}

.icon-btn:hover,
.theme-toggle:hover {
  color: var(--accent);
}

/* Grid & Cards */
.dashboard-grid {
  display: grid;
  grid-template-columns: repeat(4, 1fr);
  gap: 1.25rem;
}

.card {
  background: var(--panel);
  border-radius: var(--radius);
  padding: 1.25rem;
  border: var(--border);
}

.card h3 {
  font-size: 0.85rem;
  font-weight: 500;
  color: var(--muted);
  margin-bottom: 1rem;
}
```

```
.card-header {
  display: flex;
  justify-content: space-between;
  align-items: center;
  margin-bottom: 1rem;
}

.big-card,
.wide-card {
  grid-column: span 2;
}

.tall-card {
  grid-row: span 2;
  grid-column: 4;
}

/* Weather Card */
.big-card {
  background: linear-gradient(135deg, #0ea5e9, #38bdf8);
}

.weather-info {
  display: flex;
  justify-content: space-between;
  align-items: center;
}

.temp {
  font-size: 3.5rem;
  font-weight: 700;
  display: block;
}

.condition {
  display: inline-flex;
  align-items: center;
  gap: 0.4rem;
  background: rgba(255, 255, 255, 0.2);
  padding: 0.3rem 0.8rem;
  border-radius: 20px;
  margin-top: 0.4rem;
  font-size: 0.9rem;
}
```

```

.weather-details {
  display: flex;
  gap: 1.5rem;
  margin-top: 1rem;
}

.detail {
  text-align: center;
  font-size: 0.85rem;
}

```

Script.js

```

document.addEventListener('DOMContentLoaded', () => {
  // Mock Data unused variables removed for simplicity

  // Charts Configuration
  Chart.defaults.color = '#94a3b8';
  Chart.defaults.font.family = "'Outfit', sans-serif";

  let tempChart, rainChart, uvChart;

  const citiesData = {
    Pune: {
      temp: 28, condition: 'Clear', wind: 12, humidity: 45, uv: 7, days:
['Mon', 'Tue', 'Wed', 'Thu', 'Fri', 'Sat', 'Sun'], temps: [26, 27, 28, 29, 28, 30,
31], rain: [5, 0, 0, 10, 5, 0, 0], forecast: [
      { day: 'Mon', icon: 'fa-sun', temp: '26° / 18°' },
      { day: 'Tue', icon: 'fa-sun', temp: '27° / 19°' },
      { day: 'Wed', icon: 'fa-cloud-sun', temp: '28° / 20°' },
      { day: 'Thu', icon: 'fa-cloud-rain', temp: '29° / 21°' },
      { day: 'Fri', icon: 'fa-cloud', temp: '28° / 20°' },
      { day: 'Sat', icon: 'fa-sun', temp: '30° / 22°' },
      { day: 'Sun', icon: 'fa-sun', temp: '31° / 23°' }
    ],
  },
    Mumbai: {
      temp: 32, condition: 'Humid', wind: 15, humidity: 75, uv: 8, days:
['Mon', 'Tue', 'Wed', 'Thu', 'Fri', 'Sat', 'Sun'], temps: [30, 31, 32, 32, 33, 31,
32], rain: [20, 30, 40, 50, 20, 10, 5], forecast: [
      { day: 'Mon', icon: 'fa-cloud-sun', temp: '30° / 26°' },
      { day: 'Tue', icon: 'fa-cloud-rain', temp: '31° / 27°' },
      { day: 'Wed', icon: 'fa-bolt', temp: '32° / 27°' },
      { day: 'Thu', icon: 'fa-cloud-rain', temp: '32° / 26°' },
    ]
    }
  }

```

```

        { day: 'Fri', icon: 'fa-cloud-sun', temp: '33° / 27°' },
        { day: 'Sat', icon: 'fa-sun', temp: '31° / 26°' },
        { day: 'Sun', icon: 'fa-sun', temp: '32° / 26°' }
    ]
},
Delhi: {
    temp: 35, condition: 'Sunny', wind: 10, humidity: 40, uv: 9, days:
['Mon', 'Tue', 'Wed', 'Thu', 'Fri', 'Sat', 'Sun'], temps: [33, 35, 36, 38, 37, 39,
40], rain: [0, 0, 5, 0, 0, 0, 0], forecast: [
    { day: 'Mon', icon: 'fa-sun', temp: '33° / 22°' },
    { day: 'Tue', icon: 'fa-sun', temp: '35° / 24°' },
    { day: 'Wed', icon: 'fa-sun', temp: '36° / 25°' },
    { day: 'Thu', icon: 'fa-sun', temp: '38° / 26°' },
    { day: 'Fri', icon: 'fa-sun', temp: '37° / 25°' },
    { day: 'Sat', icon: 'fa-sun', temp: '39° / 27°' },
    { day: 'Sun', icon: 'fa-sun', temp: '40° / 28°' }
]
},
Bangalore: {
    temp: 24, condition: 'Cloudy', wind: 18, humidity: 60, uv: 6, days:
['Mon', 'Tue', 'Wed', 'Thu', 'Fri', 'Sat', 'Sun'], temps: [22, 23, 24, 25, 23, 22,
24], rain: [40, 30, 20, 10, 20, 30, 40], forecast: [
    { day: 'Mon', icon: 'fa-cloud-rain', temp: '22° / 18°' },
    { day: 'Tue', icon: 'fa-cloud', temp: '23° / 18°' },
    { day: 'Wed', icon: 'fa-cloud-sun', temp: '24° / 19°' },
    { day: 'Thu', icon: 'fa-sun', temp: '25° / 19°' },
    { day: 'Fri', icon: 'fa-cloud-sun', temp: '23° / 19°' },
    { day: 'Sat', icon: 'fa-cloud-rain', temp: '22° / 18°' },
    { day: 'Sun', icon: 'fa-cloud-sun', temp: '24° / 19°' }
]
}
};

function updateDashboard(city) {
    const data = citiesData[city];
    const citySelect = document.getElementById('citySelect');
    const selectedOption = citySelect.options[citySelect.selectedIndex];

    // Update Locations
    document.querySelector('.location h2').innerText = city;
    document.querySelector('.location p').innerText =
selectedOption.text.split(', ')[1] || 'India';

    // Update Main Weather
    document.querySelector('.temp').innerText = `${data.temp}°`;

```

```

document.querySelector('.condition span').innerText = data.condition;

// Update Details
const details = document.querySelectorAll('.detail span');
details[0].innerText = `${data.wind} km/h`;
details[1].innerText = `${data.humidity}%`;
details[2].innerText = city === 'Pune' ? '10 km' : '8 km'; // Mock
visibility

// Update UV Index
document.getElementById('uvValue').innerText = data.uv;
const uvText = data.uv > 8 ? 'Very High' : data.uv > 5 ? 'Moderate' : 'Low';
document.querySelector('.uv-value small').innerText = uvText;

// Update Forecast
const forecastList = document.querySelector('.forecast-list');
forecastList.innerHTML = '';
data.forecast.forEach(item => {
  const li = document.createElement('li');
  li.className = 'forecast-item';
  li.innerHTML = `
    <span class="forecast-day">${item.day}</span>
    <i class="fa-solid ${item.icon} forecast-icon"></i>
    <span class="forecast-temp">${item.temp}</span>
  `;
  forecastList.appendChild(li);
});

// Update Charts
tempChart.data.datasets[0].data = data.temps;
tempChart.update();

rainChart.data.datasets[0].data = data.rain;
rainChart.update();

uvChart.data.datasets[0].data = [data.uv, 12 - data.uv];
uvChart.data.datasets[0].backgroundColor[0] = data.uv > 8 ? '#ef4444' :
data.uv > 5 ? '#eab308' : '#22c55e';
uvChart.update();
}

// Chart Helper Function
function createChart(id, type, labels, label, color, bgColor, fill = false) {
  return new Chart(document.getElementById(id).getContext('2d'), {
    type: type,

```

```

    data: {
      labels: labels,
      datasets: [{
        label: label,
        data: [],
        borderColor: color,
        backgroundColor: bgColor,
        borderWidth: type === 'doughnut' ? 0 : (type === 'line' ? 3 :
0),

        borderRadius: type === 'bar' ? 4 : 0,
        tension: 0.4,
        fill: fill,
        pointBackgroundColor: '#fff',
        pointBorderColor: color,
        pointRadius: type === 'line' ? 4 : 0,
        pointHoverRadius: type === 'line' ? 6 : 0,
        cutout: type === 'doughnut' ? '80%' : undefined,
        barThickness: type === 'bar' ? 12 : undefined
      }]
    },
    options: {
      responsive: true,
      maintainAspectRatio: false,
      plugins: {
        legend: { display: false },
        tooltip: { enabled: type !== 'doughnut' }
      },
      scales: {
        y: type !== 'doughnut' ? {
          grid: { display: type === 'line', color: 'rgba(255, 255,
255, 0.05)' },

          display: type === 'line',
          suggestedMin: type === 'line' ? 15 : 0,
          suggestedMax: type === 'line' ? 45 : 100
        } : { display: false },
        x: { grid: { display: false }, display: type !== 'doughnut' }
      },
      rotation: type === 'doughnut' ? -90 : 0,
      circumference: type === 'doughnut' ? 180 : 360
    }
  });
}

// Initialize Charts using Helper

```

```

tempChart = createChart('tempChart', 'line', citiesData.Pune.days, 'Temperature (°C)', '#38bdf8', 'rgba(56, 189, 248, 0.1)', true);

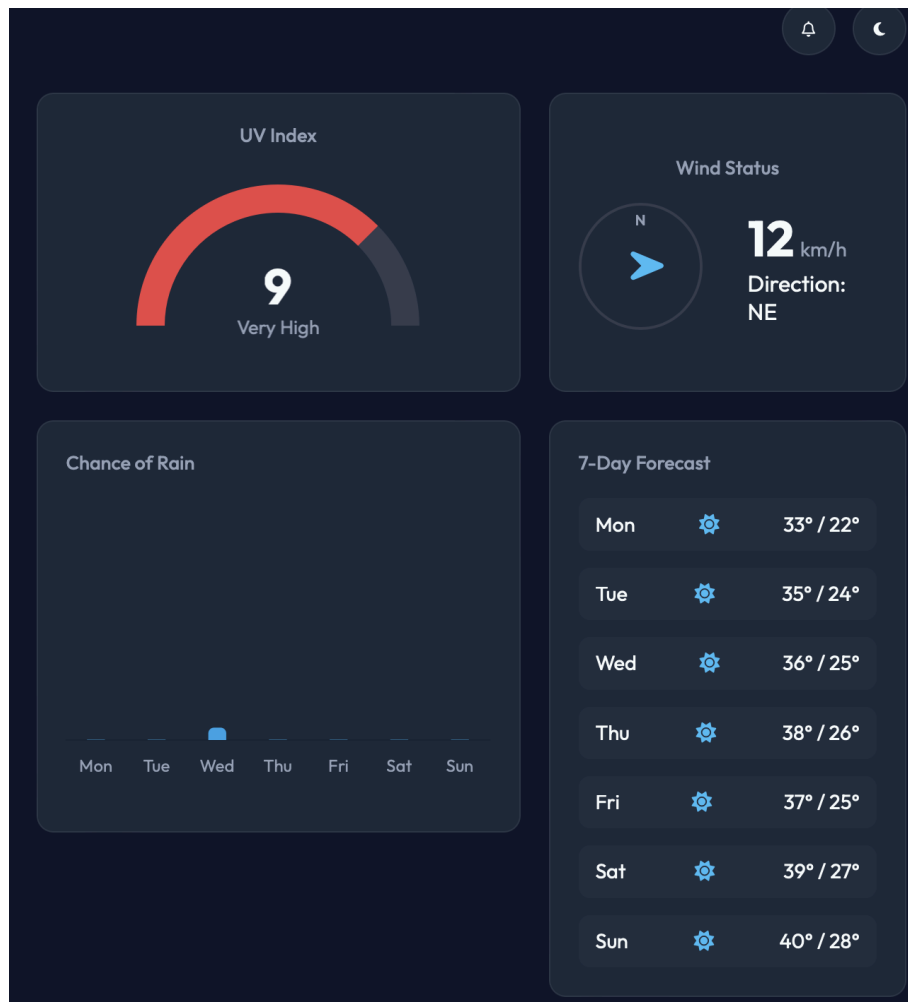
rainChart = createChart('rainChart', 'bar', citiesData.Pune.days, 'Chance of Rain (%)', undefined, '#0ea5e9');

// UV Chart
const uvCtx = document.getElementById('uvChart').getContext('2d');
uvChart = new Chart(uvCtx, {
  type: 'doughnut',
  data: {
    labels: ['UV Index', 'Remaining'],
    datasets: [{
      data: [0, 12],
      backgroundColor: ['#22c55e', 'rgba(255, 255, 255, 0.1)'],
      borderWidth: 0,
      cutout: '80%'
    }]
  },
  options: {
    responsive: true,
    maintainAspectRatio: false,
    plugins: {
      legend: { display: false },
      tooltip: { enabled: false }
    },
    rotation: -90,
    circumference: 180
  }
});

```

Output:





Conclusion

The assignment successfully demonstrated the development of an interactive data visualization dashboard using HTML, CSS, JavaScript, and visualization libraries such as Chart.js or D3.js. The project highlighted how graphical representations simplify complex data analysis and improve user understanding. This experiment strengthened skills in front-end development, data handling, and visualization techniques, which are essential in modern full stack applications.